

LAB-8 Revision Notes (JavaScript UDF, Popup Boxes & Validations)

1. User Defined Function (UDF)

Syntax

```
function functionName(parameters)
{
    // statements
    return value;
}
```

Calling Function

```
functionName(arguments);
```

Example

```
function add(a, b)
{
    return a + b;
}
```

```
let ans = add(10, 20);
```

2. Popup Boxes

A) alert()

Purpose

Displays a message.

Syntax

```
alert("Message");
```

Example

```
alert("Welcome");
```

B) prompt()

Purpose

Takes input from the user.

Syntax

```
let variable = prompt("Message");
```

For Number

```
let num = parseInt(prompt("Enter number"));
```

C) confirm()

Purpose

Returns **true** or **false**

Syntax

```
let result = confirm("Do you want to continue?");
```

3. Input Validation

Check Empty Input

```
if(value == "")
```

or

```
if(value.length == 0)
```

Check Null

(User presses Cancel)

```
if(value == null)
```

Check Number

```
isNaN(value)
```

Returns

- true → Not a Number
- false → Valid Number

Example

```
if(isNaN(num))
```

Positive Number

```
if(num > 0)
```

Negative Number

```
if(num < 0)
```

Zero

if(num == 0)

4. Calculator

Operators

+

-

*

/

%

Logic

Take Number 1

Take Number 2

Take Operator

Use switch or if-else

Display Result

5. Prime Number

Formula

A number greater than 1 having only two factors.

Logic

Input Number

Count factors

If factors == 2

Prime

Else

Not Prime

OR

Check divisibility from 2 to n-1

6. Factorial

Formula

$n!$

$5!$

$5 \times 4 \times 3 \times 2 \times 1$

$= 120$

Logic

fact = 1

Multiply from 1 to n

Output fact

7. Fibonacci Series

Formula

0 1 1 2 3 5 8 13...

Variables

first

second

next

Logic

first = 0

second = 1

next = first + second

Repeat

8. Palindrome Number

Formula

Original Number

↓

Reverse Number

↓

Compare

Equal → Palindrome

Else → Not Palindrome

Logic

Store Original Number

Reverse Digits

Compare Original and Reverse

9. Armstrong Number

Formula (3-digit)

153

$$1^3 + 5^3 + 3^3$$

$$1 + 125 + 27$$

153

Logic

Store Original Number

Take Last Digit

Cube Digit

Add

Repeat

Compare Sum and Original

10. Reverse Number Logic

Formula

digit = num % 10

reverse = reverse * 10 + digit

num = Math.floor(num / 10)

11. Extract Last Digit

digit = num % 10;

12. Remove Last Digit

num = Math.floor(num / 10);

13. Count Digits

count++;

num = Math.floor(num / 10);

14. Cube of Digit

sum = sum + digit ** 3;

or

sum = sum + Math.pow(digit, 3);

15. Common Loop Syntax

for Loop

```
for(initialization; condition; increment)
{
}
```

while Loop

```
while(condition)
{
}
```

16. Decision Making

if

```
if(condition)
{
}
```

if...else

```
if(condition)
{
}
else
{
}
```

else if

```
if(condition)
{
}
else if(condition)
{
}
else
{
}
```

switch

```
switch(expression)
{
case value:
    break;

default:
}
```

17. Output

HTML Page

```
document.write();
```

Console

```
console.log();
```

Popup

```
alert();
```

Quick Revision Table

Program	Main Formula / Logic
Calculator	+ - * / % using switch or if-else
Prime	Check divisibility from 2 to n-1 or count factors
Factorial	fact = fact × i
Fibonacci	next = first + second
Palindrome	Reverse number and compare
Armstrong	Sum of cubes (or powers) of digits
Reverse Number	reverse = reverse * 10 + digit
Last Digit	digit = num % 10
Remove Digit	num = Math.floor(num / 10)
Prompt	prompt()
Alert	alert()
Confirm	confirm()
Validation	isNaN(), null, empty string